APPLIED DATA SCIENCE FOR BUSINESS & SCIENCE · TEAM PROJECT

WP2 · Data Engineering

TEAM PROJECT · WORK PACKAGE 2

WP2 · Data engineering

Deadline · 18 November
Moodle: TEAM_XX-WP2.pdf max. 10 slides

Be creative: think beyond the dataset

Your dataset is **curated**: it is the end of a pipeline. Picture the organisation behind your use case (company, public entity or hospital) running in production: which systems generate data at each stage, how they connect, and which new sources would add value. Design **all the data pipelines** you consider the use case needs, not just one CSV.

TASKS

- **Source systems:** what each records, owner, volume, throughput, quality, refresh
- **Architecture:** open source, on-premise or cloud, justified end to end
- **Ingestion:** batch or streaming per source, justified by the decision
- **Raw layer:** storage, file formats, partitions, retention
- **ETL:** cleaning, joins, deduplication, rules that build business entities
- **Persistance:**Database model, PK/FK, indexing, upserts, data dictionary
- **Scale and infrastructure** sized to the expected throughput
- **Governance,** quality tests and known defects across all phases

PRESENTED TO

The CTO/CDO and the platform team

Cost-effective, stable, maintainable and secure with personal data.

THEY WILL ASK

1. What throughput, and which types of processes (batch, streaming)?
2. Which architectural elements are required?
3. How are the data transformed?

Questions?

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